

EXPERIMENT 2

EXPLORING THE IP

AIM:

To study different types of network cables, their categories, data rates, advantages, disadvantages, applications, and the construction of straight-through and crossover Ethernet cables.

A) STUDY OF DIFFERENT TYPES OF NETWORK CABLES

1. UNSHIELDED TWISTED PAIR (UTP) CABLE:

UTP is a twisted-pair network cable in which the individual wire pairs are not provided with additional shielding.

Categories

- Category 3 – 10 Mbps
- Category 5 – Up to 100 Mbps
- Category 5e – 1 Gbps

Advantages

- Cheaper in cost.
- Easy to install.
- Has a smaller overall diameter.
- Suitable for Ethernet networks.

Disadvantages

- More prone to electromagnetic interference (EMI) and noise.

Applications

- 10Base-T Ethernet
- Fast Ethernet
- Gigabit Ethernet

The above categories, transmission rates and applications are given in the supplied lab manual.

2. SHIELDED TWISTED PAIR (STP) CABLE:

STP is a twisted-pair cable that contains shielding to reduce the effect of electromagnetic interference and external noise.

Category

- Category 6 / 6a – Up to 10 Gbps

Advantages

- Shielded construction.
- Faster than basic UTP in the specified applications.
- Less susceptible to noise and interference.

Disadvantages

- More expensive.
- Requires greater installation effort.

Applications

- Gigabit Ethernet
- 10G Ethernet
- Data centres

The supplied manual identifies Category 6/6a STP for 10 Gbps applications and notes its shielding and reduced susceptibility to interference.

3. COAXIAL CABLE:

Coaxial cable consists of a central conductor surrounded by insulation and shielding.

Types

- RG-6
- RG-59
- RG-11

Data Transmission

- Approximately 10–100 Mbps in the applications listed in the manual.

Advantages

- High bandwidth.
- Immune to interference.
- Low signal loss.
- Versatile.

Disadvantages

- Limited distance in the stated applications.
- Cost can be higher.
- Bulky in size.

Applications

- Television networks.
- High-speed Internet connections.

These cable types and characteristics are included in the supplied practical manual.

4. FIBRE OPTIC CABLE:

Fibre optic cable transmits data using light through optical fibres.

Types

- Single Mode
- Multi Mode

Data Transmission

- Up to 100 Gbps as specified in the laboratory manual.

Advantages

- High speed.
- High bandwidth.
- High security.
- Suitable for long-distance communication.

Disadvantages

- Expensive.

- Requires skilled installers.
- The supplied manual specifies around 100 metres as the maximum distance for the stated setup.

Applications

- High-speed communication.
- Long-distance networking.
- High-bandwidth network connections.

The supplied manual lists single-mode and multi-mode fibre, along with its high-speed, high-bandwidth and long-distance characteristics.

5. TWINAXIAL CABLE:

Twinaxial cable, also called Twinax, contains two inner conductors and shielding. It is designed for high-speed data communication over relatively short distances.

Advantages

- Supports high-speed data transmission.
- Provides good protection against interference.
- Suitable for short-distance high-speed connections.
- Reliable for data-centre applications.

Disadvantages

- More expensive than ordinary twisted-pair cables.
- Limited distance compared with fibre optic cables.
- Less flexible than some other network cables.

Applications

- Data centres.
 - Server connections.
 - Storage networking.
 - High-speed short-distance communication.
-

6. RIBBON CABLE:

Ribbon cable consists of multiple parallel conductors arranged side-by-side in a flat form.

Advantages

- Multiple wires are arranged together.
- Easy to identify individual conductors.
- Compact and convenient for internal connections.
- Simple to install inside equipment.

Disadvantages

- Can experience crosstalk between adjacent conductors.
- Not suitable for long-distance network communication.
- Can become difficult to manage when many cables are used.

Applications

- Internal computer connections.
 - Networking equipment.
 - Electronic devices.
 - Data and control connections inside equipment.
-

7. TELEPHONE CABLE:

Telephone cable is commonly used for voice communication and can also support data communication in technologies such as DSL.

Advantages

- Low cost.
- Easy to install.
- Lightweight.
- Flexible.
- Suitable for telephone communication.

Disadvantages

- Lower bandwidth than modern Ethernet cables.
- More susceptible to noise and interference.
- Limited data transmission capability.

Applications

- Telephone connections.
- Traditional communication networks.
- DSL Internet connections.

B) MAKING ETHERNET CABLE

PROCEDURE:

Step 1:

Begin by threading the cable shields onto the cable.

Step 2:

Strip approximately 1.5 cm of cable shielding from both ends using the cable stripper/crimping tool.

Step 3:

Untangle the wires. There should be four twisted pairs. Arrange the wires according to the required colour arrangement.

For a crossover cable, one end follows arrangement A and the other end follows arrangement B.

Step 4:

Arrange all the wires in a straight line and cut them evenly. Insert the wires carefully into the RJ45 connector without changing their order.

The supplied manual instructs the user to hold the RJ45 plug with the clip side facing away and the gold pins facing towards the user while inserting the wires.

Step 5:

Push the wires completely into the RJ45 connector. Make sure the cable shielding reaches the required position inside the connector.

Step 6:

Insert the RJ45 connector into the crimping tool and press it carefully to secure the wires and connector.

Step 7:

Repeat the process for the other end.

- For a **crossover cable**, use arrangement B at the second end.

- For a **straight-through cable**, use arrangement A at the second end.

RESULT

Thus, the different types of network cables were studied, and the construction of **straight-through and crossover Ethernet cables** using RJ45 connectors was understood.

